Ex. No: 10

Title: L2, Dropout and Early Stopping Comparison

Packages: TensorFlow, Keras, NumPy, Matplotlib

AIM:

To compare the effect of L2 regularization, Dropout and Early Stopping on a neural network.

ALGORITHM:

1. Start the program and import the required libraries.
2. Load the digits dataset, scale it and split it into training and testing sets.
3. Create a function to build a neural network with optional L2 regularization and Dropout.
4. Create four models: Baseline, L2, Dropout and Early Stopping.
5. Train each model and store its validation loss.
6. Evaluate each model on the test data and print the accuracy.
7. Plot the validation loss of all models for comparison.
8. Display the output and stop.

PROGRAM:

```
import matplotlib.pyplot as plt
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from tensorflow import keras
from tensorflow.keras import layers, regularizers

X, y = load_digits(return_X_y=True)
X = X / 16.0
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=0)

def build(reg=None, drop=0.0):
    return keras.Sequential([
        layers.Input(shape=(64,)),
        layers.Dense(128, activation="relu", kernel_regularizer=reg),
        layers.Dropout(drop),
        layers.Dense(128, activation="relu", kernel_regularizer=reg),
        layers.Dropout(drop),
        layers.Dense(10, activation="softmax")
    ])

early = keras.callbacks.EarlyStopping(monitor="val_loss", patience=3,
restore_best_weights=True)

experiments = {
    "Baseline": (build(), []),
    "L2": (build(reg=regularizers.l2(0.01)), []),
    "Dropout": (build(drop=0.3), []),
    "Early Stopping": (build(), [early]),
}

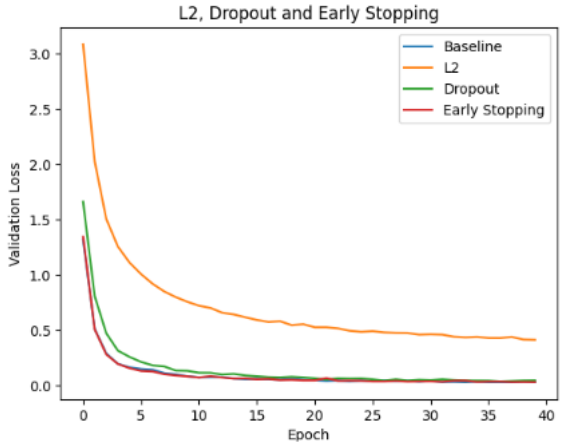
for name, (model, callbacks) in experiments.items():
    model.compile(optimizer="adam", loss="sparse_categorical_crossentropy",
metrics=["accuracy"])
```

```
    history = model.fit(X_train, y_train, epochs=40, validation_split=0.2,
                        callbacks=callbacks, verbose=0)
    acc = model.evaluate(X_test, y_test, verbose=0)[1]
    print(name, "- Test Accuracy =", round(acc, 4))
    plt.plot(history.history["val_loss"], label=name)

plt.xlabel("Epoch")
plt.ylabel("Validation Loss")
plt.title("L2, Dropout and Early Stopping")
plt.legend()
plt.show()
```

OUTPUT:

```
Baseline - Test Accuracy = 0.9722
L2 - Test Accuracy = 0.9519
Dropout - Test Accuracy = 0.9722
Early Stopping - Test Accuracy = 0.9704
```



RESULT:

L2 regularization, Dropout and Early Stopping were compared successfully. These techniques reduce overfitting and keep the validation loss under control.